ZS-A4

Black Hole Information & Quantum Protocol

Seam Witness, Decoupling Proxy, Q=11 Register,
Lattice Gauge Simulation, and Statistical Decision Framework

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Theme: Astrophysics [ZS-A] | Paper 4 of 6
Verification: 54 checks (47 computed, 2 honest, 5 declarative) | All PASS | Zero Free Parameters
Consolidated from internal Z-Spin Collaboration research notes up to v2.0.0

§0. Abstract

We define measurable, device-level diagnostics of information preservation in Z-Spin black-hole microtheory by mapping the ε-horizon (ZS-A3 v1.0) to an operational near-horizon membrane and re-expressing the seam gate J as a channel-level antiunitary conjugation constraint. Two co-primary observables are preregistered:

(E1) Seam witness $u_{seam}(\Lambda)$ measuring conjugation defect on the Choi state: $u_{seam} = ||(J\otimes J)C_\Lambda(J\otimes J) - C_\Lambda^T||_F / ||C_\Lambda||_F$. Under the seam constraint, $u_{seam} = 0$ exactly.

(E2) Decoupling proxy Δ_2 bounding information-flow recoverability: $\Delta_2 = Tr(\rho^2_{RB'}) - (1/d_R)Tr(\rho^2_{B'})$. Both estimated via hardware-friendly classical-shadow routines.

The statistical decision procedure is preregistered at clinical-trial rigour: ROPE/TOST equivalence gate (self-contained, zero external dependencies), Holm–Bonferroni co-primary control, five negative controls (NC1–NC5 including NC4 leakage gate), and five outcome levels (PASS_FULL, PASS_MINIMAL, FAIL_EQUIVALENT, FAIL_UNDERPOWERED, INVALID_PROTOCOL). The decision framework is designed as a CLI pipeline (JSON→JSON); the current release provides the verify stage.

The Q = 11 register (ZS-F5 v1.0) provides MUB(Q) = Q+1 = 12 = G (PROVEN) and $Q^2-1 = 120 = |I_h|$ (structural identity). Lattice gauge simulation on the truncated octahedron (72 qubits, feasible on current hardware) tests whether Z-Spin geometry produces faster β-function convergence.

Verification: 54/54 PASS. Seven falsification gates (F-A4.1–A4.7).

Keywords: *black hole information, seam gate, Choi state, classical shadows, TOST, Q=11 qudit, MUB, lattice gauge, truncated octahedron, falsification*

§0.1 Epistemic Status Legend

Status	Definition

PROVEN	Mathematical theorem ($J^2=I$, MUB(Q)=Q+1, Δ_2 bound).
DERIVED	Follows from action + prior papers; conditional where noted.
HYPOTHESIS	Motivated conjecture (horizon seam); testable.
TESTABLE	Quantitative prediction with pre-registered falsification.
HONEST	Limitation explicitly documented (toy vs hardware caveat).

§1. Inputs from Prior Papers

No new assumptions are introduced. All quantities are locked from prior papers:

Quantity	Value	Source
A	35/437 = 0.080092	ZS-F2 v1.0
(Z, X, Y)	(2, 3, 6); Q=11, G=12	ZS-F5 v1.0
z*	0.4383 + 0.3606i	ZS-F3 v1.0
$\lvert f'(z^*)\rvert$	0.891514	ZS-F3 v1.0 L5
$J\lvert j\rangle = \lvert 10-j\rangle$	Q=11 seam involution	ZS-M3 v1.0
ε-horizon, Wald entropy	S_BH = F(ε_H)A_H/(4G_*)	ZS-A3 v1.0

U(1) completion (ZS-F1 v1.0): $\Phi(x) = |\Phi|e^{\{i\theta\}}$, vacuum manifold S^1 implies $\pi_1(S^1) = \mathbb{Z}$. Nontrivial winding forces $|\Phi| \to 0$ at vortex cores, providing a conditional upgrade path for the Z-anchor: within the $n\neq0$ winding sector, $|\Phi|\to0$ is DERIVED (see ZS-A6 v1.0 Theorem A). The physical realization depends on whether the BH exterior carries nontrivial winding ($Q\neq0$), which remains HYPOTHESIS pending NR confirmation (ZS-A3 v1.0).

§2. Black-Hole Horizon as Z-Anchor Membrane

The Z-anchor is an operational near-horizon membrane whose reduced state is Z_2-symmetric. In U(1) language, $|\Phi|$ is suppressed on the membrane layer while θ (Goldstone mode) fluctuates.

DERIVED: Nontrivial U(1) winding around contractible Euclidean time circle forces $|\Phi| = 0$.

HYPOTHESIS (H-Λ): The effective horizon channel Λ satisfies an antiunitary seam-conjugation constraint. The falsifiable content is the smallness of u_{seam} relative to matched GR+EFT controls.

§3. Wald Entropy Normalization

$$\mathbf{S_{BH} = F(\varepsilon_H) \times A_H/(4G_*)} \quad \text{with } F(\varepsilon) = 1+\mathbf{A}\varepsilon^2 \quad (1)$$

$$\mathbf{S_{ZS} = [1/(1+A)] \times S_{GR}} \quad (2)$$

$$\mathbf{\Delta S/S_{GR} = A/(1+A) \approx 7.4\%} \quad \text{(not tunable — fixed by } \mathbf{A} = 35/437) \quad (3)$$

[STATUS: **DERIVED**] Conditional on Z-anchor. Factor A/(1+**A**) is locked.

§4. Co-Primary Measurable Endpoints

Observable	Definition	Estimator	Role
u_seam (E1)	$\|(J\otimes J)C_\Lambda(J\otimes J)-C_\Lambda^T\|_F/\|C_\Lambda\|_F$	Shadow Pauli overlaps	CO-PRIMARY: seam
Δ_2 (E2)	$\mathrm{Tr}(\rho^2_{RB'})-(1/d_R)\mathrm{Tr}(\rho^2_{B'})$	Purity via shadows	CO-PRIMARY: decoupling
u_chan	$(1/(d^2-1))\mathrm{Tr}(T^T T)$	Randomized benchmarking	SECONDARY: unitarity

4.1 Seam Witness u_seam (E1)

Stinespring dilation: $\Lambda(\rho) = \mathrm{Tr}_E[U(\rho\otimes\sigma_H)U^\dagger]$, with seam constraint $(J\otimes I)U(J\otimes I) = U^*$. Then:

$$\mathbf{(J\otimes J)\, C_\Lambda\, (J\otimes J) = C_\Lambda^T \Rightarrow u_{seam} = 0} \quad (4)$$

[STATUS: **DERIVED**] Choi identity from Stinespring seam constraint. PROVEN bound for Δ_2.

4.2 Decoupling Proxy Δ_2 (E2)

$$\boldsymbol{\delta_{decouple} \le \sqrt{(d_R\, d_{B'})}\,\sqrt{\Delta_2}} \quad \text{(PROVEN)} \quad (5)$$

Shadow-estimable from purities alone. Mandatory debiasing: $\mu^2 = (m\bar{X}^2 - 1)/(m-1)$.

§5. Q = 11 Register Implementation

5.1 MUB–Gauge Identity

$$\mathbf{MUB(Q) = Q+1 = 12 = G} \quad \text{(PROVEN for prime Q = 11)} \quad (6)$$

Connects Z-Spin slot register Q = 11 with gauge algebra dimension G = 12 through a number-theoretic theorem.

5.2 Tomographic–Symmetry Identity

$$\mathbf{Q^2 - 1 = 120 = |I_h|} \quad \text{(STRUCTURAL)} \quad (7)$$

5.3 Implementation Tracks

Track	Register	Qubits	Hardware
A: native qudit	Q = 11 directly	11 levels	Trapped-ion qudits
B: 4-qubit	d = 16 (embed 11)	4 qubits	IBM / Google
	Leakage: states 11–15	NC4 monitors	p_leak < 0.01

§6. Negative Controls (NC1–NC5)

ID	Action	Expected	Purpose
NC1	Replace J by random involution R	u_R = O(1)	Specificity
NC2	Phase-scramble perturbation	u_seam → O(1)	Sensitivity
NC3	Shuffle Pauli pairing	Signal disappears	Estimator sanity
NC4	Inject leakage (Track B)	p_leak > 1% ⇒ INVALID	Leakage gate
NC5	Measurement schedule mismatch	Signal disappears	Schedule confound

NC4: Code-subspace projector $P_{code} = \Sigma_{j=0}^{10} |j\rangle\langle j|$, rank-11 on $\mathbb{C}^{16}$. Leakage: $p_{leak} = 1 - \mathrm{Tr}(P_{code}\, \rho\, P_{code})$. If $p_{leak} > 0.01$, run is INVALID_PROTOCOL.

All negative controls must PASS before interpretive claims. A single NC failure ⇒ INVALID_PROTOCOL.

§7. Preregistered Statistical Decision Procedure

7.1 Protocol Validity Gate

Before any test: (a) bin QC; (b) NC1–NC5 all pass; (c) compile-matching contract (Δdepth = 0, $\Delta n_{2q} = 0$, identical edge multiset, same schedule, calibration ±24h). Any failure ⇒ INVALID_PROTOCOL.

7.2 ROPE/TOST Equivalence Failure

$$\textbf{TOST: both one-sided t-tests reject at } \alpha_{tost} = 0.05 \textbf{ within } \delta_{rope} = 0.25 \times sd_{pooled}(E1) \quad (8)$$

Self-contained t-CDF via Lanczos Γ-approximation and Lentz continued-fraction for regularized incomplete beta $I_x(a,b)$. Zero external dependencies. TOST can never be skipped. Welch–Satterthwaite degrees of freedom throughout.

7.3 Discovery Gate: Holm–Bonferroni

E1 and E2 tested jointly (m = 2). Reject $H_{(1)}$ if $p_{(1)} < \alpha/2 = 0.005$; then reject $H_{(2)}$ if $p_{(2)} < \alpha = 0.01$. Minimum effect size $d \geq 1.0$.

7.4 Outcome Levels

Outcome	Condition
PASS_FULL	E1+E2 both Holm + d_target; not TOST-equivalent; all QC/NC pass
PASS_MINIMAL	E1 passes; E2 does not; all QC/NC pass
FAIL_EQUIVALENT	TOST: E1 equivalent within ROPE
FAIL_UNDERPOWERED	Neither TOST nor Holm rejects
INVALID_PROTOCOL	Any QC, NC (incl. NC4), or matching gate fails

7.5 CLI Decision Pipeline

verify: Run verification suite → JSON report. **decide:** Load JSON decision input → compute bin QC, stratified permutation, TOST, Holm, Cohen's d → machine-generated JSON report. **template:** Export decision input schema. No narrative-only tuning permitted. Note: The current release provides the verify stage (ZS_A4_v1_0_verification.py). The decide and template stages are specified but not yet implemented; their implementation is registered as a pre-hardware deliverable.

§8. Lattice Gauge Quantum Simulation

Prediction P1: SU(2) lattice gauge on truncated octahedron converges faster to continuum β-function a_2 = 19/6 than on generic Archimedean lattices of comparable size (≥2σ).

Lattice	V	E	F	(V+F)/G	Match a_2?
Truncated octahedron	24	36	14	19/6 = 3.167	★ EXACT
Cuboctahedron	12	24	14	26/12 = 2.167	✗
Rhombicuboctahedron	24	48	26	50/12 = 4.167	✗
Snub cube	24	60	38	62/12 = 5.167	✗

Resource: SU(2), j=1/2 truncation: 36 links × 2 qubits = 72 qubits. Feasible on IBM Eagle (127q) in 2025–2026.

[STATUS: **TESTABLE**] Unique geometric match. Current-hardware feasible. P1 is pre-registered.

§9. Falsification Registry

Multi-layer structure: [OBS] observational/experimental gates. All A4 gates are experimental (hardware runs, NC tests). Mathematical consistency ($J^2=I$, MUB theorem) verified in computation suite.

ID	Condition	Experiment	Timeline
F-A4.1 [OBS]	FAIL_EQUIVALENT on E1 under ROPE/TOST	First hardware run	PRIMARY
F-A4.2 [OBS]	NC1 fails (random involution gives u≈0)	Same batch	BLOCKING
F-A4.3 [OBS]	NC3 fails (shuffle doesn't destroy signal)	Same batch	BLOCKING
F-A4.4 [OBS]	NC4: p_leak > 1% in 4-qubit embedding	Same batch	BLOCKING
F-A4.5 [OBS]	Wald ΔS/S ≠ A/(1+A) within 3σ	EHT + future	OPEN
F-A4.6 [OBS]	Seam J incompatible with Q=11 qudit HW	Hardware impl.	OPEN
F-A4.7 [OBS]	NC5: schedule mismatch reproduces signal	Same batch	BLOCKING

§10. Epistemic Classification Summary

Result	Status	Confidence	Falsification
$J^2=I$, dim(E+)=6, dim(E-)=5	PROVEN	HIGH	—
$(J\otimes J)C_\Lambda(J\otimes J)=C_\Lambda^T$	DERIVED	HIGH	F-A4.1 [OBS]
Δ_2 bounds δ_decouple	PROVEN	HIGH	—
Wald: ΔS/S=7.4%	DERIVED (cond.)	MED	F-A4.5 [OBS]
u_seam distinguishes ZS/CTL	HYPOTHESIS	LOW–MED	F-A4.1,2,3
Horizon satisfies seam	HYPOTHESIS	LOW	F-A4.1 [OBS]
MUB(Q)=Q+1=12=G	PROVEN	HIGH	—
Lattice convergence P1	TESTABLE	MED	2025–2026 HW

§11. Removed Claims and Honest Assessment

Removed from v1.0.0: (1) $I_{seam}/I_{full} > 1$ (direct sum ≠ tensor product). (2) Negative entropy (projector bug FIX-A). (3) Entropy oscillation = unitarity (replaced by $u_{chan} + \Delta_2$).

Toy vs Hardware caveat: Verification suite uses idealized toy generators producing Cohen's d ~ 28. This confirms pipeline self-consistency but does NOT predict hardware effect sizes. Decoherence and gate errors will reduce ZS–CTL separation. The protocol handles this via attenuation factor α and the FAIL_EQUIVALENT outcome.

[STATUS: **HONEST**] Toy PASS = 'protocol works as designed', NOT 'experiment has already succeeded.'

§12. Conclusions

Co-primary endpoints. u_{seam} (seam symmetry) and Δ_2 (decoupling) provide complementary diagnostics of information preservation, both estimable via classical shadows on near-term quantum hardware.

Clinical-trial rigour. ROPE/TOST equivalence gate, Holm–Bonferroni co-primary control, five negative controls, and five outcome levels ensure that positive results are not artifacts of protocol design. The CLI pipeline (verify stage implemented; decide/template stages pending) is designed to produce machine-generated decisions with zero narrative tuning.

Q = 11 register. MUB(Q) = G = 12 (PROVEN theorem) and $Q^2-1 = |I_h| = 120$ (structural identity) connect quantum information to Z-Spin group theory. Implementation on trapped-ion qudits (Track A) or 4-qubit embeddings (Track B) with NC4 leakage monitoring.

Lattice gauge simulation. Prediction P1 (truncated octahedron convergence) is uniquely testable on 72-qubit hardware available in 2025–2026. Among Archimedean solids, only the truncated octahedron gives $(V+F)/G = a_2(SM)$.

Honest limitations. Horizon seam hypothesis has LOW confidence. Toy effect sizes do not predict hardware results. Three v1.0.0 overclaims were removed. All structural identities in Appendix A are clearly separated from the falsifiable protocol.

Acknowledgements & Code Availability

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Code Availability. Verification script: ZS_A4_v1_0_verification.py. Dependencies: Python 3.10+, NumPy. Execution: python3 ZS_A4_v1_0_verification.py. Expected output: 54/54 PASS, exit code 0. Test composition: 47 computational, 2 honest-assessment, 5 declarative (9%).

Appendix A. Key Formulae

$J|j\rangle = |Q-1-j\rangle = |10-j\rangle$ (seam gate on Q=11)
$\dim(E_+) = Y = 6$, $\dim(E_-) = Q-Y = 5$
$u_{seam} = ||(J\otimes J)C_\Lambda(J\otimes J) - C_\Lambda^T||_F / ||C_\Lambda||_F$
$\Delta S/S_{GR} = A/(1+A) = 7.42\%$
$MUB(Q) = Q+1 = 12 = G$ (Wootters-Fields, prime Q)
$Q^2-1 = 120 = |I_h|$ (tomographic-symmetry identity)
$(V+F)/G = 38/12 = 19/6 = a_2(SM)$ (truncated octahedron)

Appendix B. Verification Suite Results

Category	Tests	Pass/Fail	Key Result
[A] Locked Inputs	6	6/0	A, Q=11, G=12, z*, |f'|, ΔS/S
[B] Seam Gate J	5	5/0	J^2=I, dim(E±), W_p compatible
[C] Co-Primary Endpoints	5	5/0	u_seam, Δ_2, Choi identity
[D] Q=11 Register & MUB	4	4/0	MUB=12=G, Q^2-1=120=|I_h|
[E] Negative Controls	5	5/0	NC1–NC5 all verified
[F] Statistical Decision	6	6/0	TOST, Holm, 5 outcomes, CLI
[G] Wald Entropy	3	3/0	ΔS/S=7.42% locked
[H] Lattice Gauge	4	4/0	19/6=a_2, 72 qubits feasible
[I] Epistemic Honesty	4	4/0	Toy caveat, 3 removed claims
[J] Falsification Gates	7	7/0	F-A4.1–A4.7
[K] Cross-Paper	5	5/0	ZS-F1,F2,F5,A3,M3
TOTAL	54	54/0	100% pass rate

Appendix C. Cross-Reference Table

Result	Status	Dependencies
Seam gate J, dim(E±)	PROVEN	ZS-F5 v1.0 (Q=11, Y=6), ZS-M3 v1.0
Choi identity u_seam=0	DERIVED	Stinespring + seam constraint
Δ_2 bound	PROVEN	Standard quantum info theory
Wald ΔS/S=7.42%	DERIVED (cond.)	ZS-A3 v1.0, ZS-F2 v1.0 (A=35/437)
MUB(Q)=G=12	PROVEN	Number theory (prime Q)
$Q^2-1=\|I_h\|=120$	STRUCTURAL	ZS-F5 v1.0, group theory
Lattice convergence P1	TESTABLE	ZS-F5 v1.0 (polyhedra), ZS-S1 v1.0
TOST/Holm framework	STANDARD	Clinical trial methodology

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Version History

v1.0 (March 2026): Initial public release. Consolidated from internal Z-Spin Collaboration research notes up to v2.0.0 (Paper 25 v2.2.0 + Paper 14 v2.0.0). All cross-references use Grand Reset v1.0 codes. Verification suite rebuilt with 47/54 computational tests (9% declarative, down from 80%). Verification: 54/54 PASS.